FINAL YEAR PROJECT PROPOSAL | BACHELOR'S DEGREE IN INFORMATION TECHNOLOGY

Amani Line

A trusted, multilingual, low-bandwidth channel for verified community alerts and local dispute resolution.

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Supervisor: _____

1. Executive Summary

In many Kenyan neighbourhoods, the first account of a tense event is a forwarded WhatsApp message: a rumoured eviction, a contested water point, a claim about a nearby clash. Unverified information spreads faster than any official correction, and small frictions escalate into disputes. The people who most need accurate information often have the least data, the weakest connectivity, and speak Swahili, Sheng or a mother tongue rather than English.

Amani Line lets residents report or query local information by **SMS, USSD, WhatsApp or a lightweight web app**. AI helps with the slow, repetitive work: understanding messages in Swahili, English and Sheng, grouping duplicates, and drafting plain-language summaries. **Humans decide what is true.** Reports are checked by a ladder of trusted local validators (ward administrators, community health volunteers, faith and youth leaders, Red Cross volunteers) and published as short bulletins showing their status (*Unverified, Being checked, Confirmed, Debunked*), their source, and a clear next step.

The same pipeline routes everyday disputes (water queues, boundary lines, rent, market stalls) to trained community mediators with a neutral, structured first response, so disagreements are settled before they become conflicts.

> **Project outcome:** a working prototype with an SMS/USSD-first flow, an AI triage layer, a validator dashboard and a verified-bulletin feed, tested with simulated and pilot users in two neighbourhoods over 12 weeks, together with a final report, source code and user documentation for academic assessment.

2. Problem Statement & Background

2.1 The problem

Residents of dense, fast-growing settlements lack a **trusted, fast, local** way to confirm what is happening around them. Official channels are slow and often not local or in the right language. Social media is quick but unverifiable. Three harms recur:

- **Rumour-driven escalation.** False or exaggerated alerts trigger panic, retaliation or avoidable displacement, especially around elections, evictions and resource shortages.
- **Slow crisis coordination.** During floods, fires or unrest, responders, chiefs and volunteers duplicate effort because there is no shared, verified picture.
- **Unresolved everyday disputes.** Minor conflicts over water, boundaries, rent or noise have no neutral first stop, so they drift toward police, vigilantism or silence.

2.2 Who is affected

User	Need	Constraint
Residents (youth, older adults, women traders)	Know what is true and what to do	Feature phones, prepaid data, low literacy for some
Local validators	A fast, credible way to confirm or correct	Limited time; fear of being blamed
Mediators and chiefs' offices	Structured, documented referrals	No case-tracking tools
Responders (Red Cross, county)	A verified, de-duplicated situation view	Noisy, unstructured inputs

2.3 Why now

Mobile money and SMS reach nearly every household, so the channels already exist. Modern language models can now handle Swahili and code-switched text well enough to *assist* triage, though not to judge truth. That gap, helping humans verify faster rather than replacing them, is where this project sits.

2.4 Existing approaches and gaps

Fact-checking outlets verify national claims, not hyperlocal ones. Crisis-mapping tools (e.g. Ushahidi-style platforms) collect reports but need heavy moderation. Mediation services exist but are offline and under-resourced. No widely used tool combines **low-bandwidth access, local validation, multilingual AI assistance and dispute routing**.

Note: a short literature review (Weeks 1–2) will confirm statistics, related systems and partner names, and all sources will be cited in the final report.

3. Project Objectives & Scope

3.1 Objectives

1. **Understand** how information and disputes flow in two pilot neighbourhoods through at least 20 interviews with residents, validators and mediators.
2. **Build** a prototype that accepts reports in Swahili, English and Sheng over SMS/USSD and WhatsApp, and publishes verified bulletins.
3. **Verify** reliably: every published bulletin carries a status, source and timestamp; AI never publishes without human approval.
4. **Resolve** disputes: route at least 80% of test dispute reports to the right mediator category with a clear next step.
5. **Evaluate** usability, trust and accessibility with real users, and document requirements, design and results for final year assessment.

3.2 Success measures

Under 30 min	At least 85%	At least 70%	Zero
median time from report to first status (test)	triage accuracy on a labelled Swahili/English set	of pilot users say they would trust a "Confirmed" bulletin	unreviewed AI-generated bulletins published

3.3 Core design principles

- **Trust and verification:** visible status ladder, named validator tier, and a "why we think this" line. Debunks get the same prominence as alerts.
- **Low bandwidth:** SMS/USSD first; the web app is an offline-capable PWA under 150 KB per page.
- **Accessibility:** short sentences, icons plus text, voice-note intake (WhatsApp) transcribed for validators, large-text mode.
- **Privacy:** phone numbers hashed, no names in public bulletins, reporter anonymity by default, data minimisation and a retention limit.
- **Multilingual and local:** Swahili and English at launch, Sheng-aware matching, local place names and landmarks rather than coordinates.
- **Clear next steps:** every bulletin ends with an action (avoid area X, contact mediator Z).

3.4 Scope

In scope	Out of scope
Intake via SMS/USSD simulator, WhatsApp sandbox and PWA AI triage: language detection, category, duplicate clustering, draft summary Validator dashboard and trust ladder Verified bulletin feed (SMS digest and web) Dispute routing to mediators with structured intake Pilot in two neighbourhoods	National-scale deployment Automated truth judgement by AI Law-enforcement integration or surveillance features Payment features Legal arbitration

3.5 Risks and safeguards

- **Misuse to spread accusations:** reports naming individuals are blocked from publication and escalated privately.
- **Validator bias or capture:** two independent validators for high-risk reports; balanced validator panels; public audit log.
- **AI errors in Sheng or dialects:** confidence thresholds; low-confidence items go to a human, never auto-published.
- **Reporter safety:** anonymity by default; no precise location for sensitive incidents.

4. Methodology & Timeline

4.1 Approach

An iterative, human-centred software development life cycle (requirements, design, build, test) in four phases, with users involved at each stage. Each phase produces documents that feed directly into the final project report.

Phase	Activities	Output
1. Discover (Wk 1–3)	Interviews, rumour and dispute mapping, validator recruitment, ethics and consent plan	Personas, journey maps, requirements
2. Design (Wk 3–5)	Trust ladder, bulletin templates in Swahili/English, SMS/USSD menus, wireframes, paper tests	Validated flows and content design
3. Build (Wk 5–9)	Next.js PWA, Supabase/PostgreSQL data layer, **FastAPI** service for intake and AI triage with human-in-the-loop, SMS gateway sandbox, WhatsApp intake, validator dashboard	Working prototype
4. Test and Report (Wk 9–12)	Usability and trust testing, accuracy evaluation, privacy review, final report and demo	Evidence, demo, roadmap

4.2 Architecture in brief

Message in (SMS/USSD/WhatsApp/PWA) to a **FastAPI** gateway, then language detection and normalisation, AI classification (rumour, incident, dispute, query) and duplicate clustering, then a draft summary. The draft enters the **validator queue**; after a human verdict the bulletin is published with its status and next step, and sent as an SMS digest and web feed. Disputes branch to a mediator queue with a structured intake form. Every step is logged for audit.

4.3 Gantt chart (12 weeks)

Task	1	2	3	4	5	6	7	8	9	10	11	12
Research and interviews	■	■	■									
Validator recruitment		■	■	■								
Content and flow design			■	■	■							
Backend and intake					■	■	■	■				
AI triage						■	■	■	■			
Validator dashboard							■	■	■			
Pilot and testing									■	■	■	
Report and demo											■	■
Milestones			■		■				■			■

Milestones (orange): Wk 3 requirements signed off | Wk 5 design validated | Wk 9 prototype complete | Wk 12 final demo.

4.4 Evaluation methods

A labelled set of about 300 Swahili/English/Sheng messages measures triage accuracy. Usability tests with 12–15 participants (including feature-phone users and older adults) measure task success and comprehension. A short trust survey compares reactions to each bulletin status. A privacy and threat review closes the build phase.

5. Budget & Resource Allocation

Estimates for the 12-week project, in Kenya shillings (KSh) at roughly KSh 130 to USD 1; to be refined with quotes. The build is done by the student developer, so cash costs are modest and mostly cover field work and services.

Category	Item	Est. cost (KSh)
People	Local validator and mediator stipends (10 people x 8 weeks)	104,000
People	Interview and usability incentives (about 40 participants, airtime/mobile money)	39,000
People	Swahili/Sheng language reviewer and content editor	52,000
Technology	SMS/USSD gateway sandbox and test credits	32,500
Technology	Hosting for FastAPI service, Next.js app and Supabase database	19,500
Technology	AI model API usage and AI development tools	39,000
Field	Transport and venue for two community sessions	32,500
Governance	Privacy review and ethics consultation	19,500
Contingency	About 10%	34,000
Total		**KSh 372,000**

Resources

- **Team:** student developer (full-stack/AI); 1 university supervisor; 1 community liaison; 2 pilot-site validator leads; community partners as advisors.
- **Partners (to approach):** ward administration, community health volunteers, local mediation committees, and a humanitarian or civic-tech organisation for guidance.
- **Tools:** Next.js, Node.js, TypeScript and Supabase/PostgreSQL for the app and data; **FastAPI (Python)** for the intake and AI triage service; an SMS/USSD provider sandbox; a multilingual LLM API; open-source translation and language-ID libraries.

6. Qualifications & Conclusion

6.1 Qualifications

The project lead is a final year Information Technology student and software engineer based in Nairobi with a background across full-stack development, data analysis and machine learning. A degree in Information Technology (Mount Kenya University) provides the systems foundation, and certifications in AWS AI, AWS Cloud Practitioner and Data Science Foundations support the cloud and AI work. Freelance delivery with Next.js, Node.js, Supabase, PostgreSQL and TypeScript matches the proposed stack, and computer-vision work with OpenCV shows applied ML experience. Code is on GitHub (github.com/Kertich).

Skill needed	Evidence
Full-stack prototyping	Freelance Next.js / Supabase / TypeScript projects
AI / ML triage and evaluation	AWS AI certification, data science training, OpenCV project
Systems design and documentation	UML and structured design coursework
Community and ethics expertise	University supervisor, ethics approval and community partners

6.2 Conclusion

Amani Line targets a practical, everyday cause of instability: people cannot quickly tell what is true nearby, and have nowhere neutral to take a dispute. By keeping humans in charge of verification, meeting people on the channels they already use, and designing for privacy and low bandwidth from day one, the project tests whether trusted local information can reduce panic, speed coordination and settle disputes earlier.

> **Why develop it further:** if the pilot shows faster, more trusted verification and successful dispute referrals, the next step is a three-county pilot with a formal partnership, a published validator code of conduct, and an open-source release so other communities can adapt it.